

ODD 2023
Tutorial Sheet – 12
Structures & Union

Software Development Fundamentals – I (15B11CI111)

Course Outcomes (CO)	
CO1	Explain various phases of software development life cycle
CO2	Explain various data types, memory allocation schemes, precedence of arithmetical and logical operations, and need of array, and structures
CO3	Draw the flow chart and write the high level code for different problems
CO4	Apply and implement functions with or without pointers for different problems
CO5	Demonstrate and implement various operations like traverse, insertion, deletion, etc. on files

Note: Students are advised to submit their solutions to their respective tutorial faculty.

Q1) [CO4] What is the output of the following code?

```
#include<stdio.h>
#include<conio.h>
void main()
{
    struct testing
    {int a;
    char b;
    float c;
    } var;
    clrscr();
    printf("\nsizeof(var) is %d",sizeof(var));
    printf("\nsizeof(var.a) is %d",sizeof(var.a));
    printf("\nsizeof(var.b) is %d",sizeof(var.b));
    printf("\nsizeof(var.c) is %d",sizeof(var.c));
    var.a=10;
    printf("\nvalue of var.a is %d",var.a);
    var.b='b';
    printf("\nvalue of var.b is %c",var.b);
    var.c=15.55;
    printf("\nvalue of var.c is %f",var.c);
    printf("\nvalue of var.a is %d",var.a);
    printf("\nvalue of var.b is %c",var.b);
    printf("\nvalue of var.c is %f",var.c);
    getch();
}
```

Q2)[CO4] What is the output of the following code?

```
#include<stdio.h>
#include<conio.h>
void main()
{
union testing
{int a;
char b;
float c;
} var;
clrscr();
printf("\nsizeof(var) is %d",sizeof(var));
printf("\nsizeof(var.a) is %d",sizeof(var.a));
printf("\nsizeof(var.b) is %d",sizeof(var.b));
printf("\nsizeof(var.c) is %d",sizeof(var.c));
var.a=10;
printf("\nvalue of var.a is %d",var.a);
var.b='b';
printf("\nvalue of var.b is %c",var.b);
var.c=15.55;
printf("\nvalue of var.a is %f",var.c);
printf("\nvalue of var.a is %d",var.a);
printf("\nvalue of var.b is %c",var.b);
printf("\nvalue of var.c is %f",var.c);
getch();
}
```

Q3) [CO4] Write a C program to accept details of 'n' employee(eno, ename, salary) and display the details of employee having highest salary. Use array of structure.

Q4)[CO4] Write a C program to create one Structure student (with members as name, rollno, and percentage) and declare it inside union. Then accept values for structure members and display them.

Q5) [CO4] Write a C program to create a structure "telephone": which is made up of a string (that is used to hold the name of the person) and an integer (that is used to hold the telephone number).